

- 1 (a) (i) angle (subtended) at centre of circle  
by an arc equal in length to the radius (of the circle) B1  
B1 [2]
- (ii) angle swept out per unit time / rate of change of angle  
by the string M1  
A1 [2]
- (b) friction provides / equals the centripetal force B1  
 $0.72 W = m d \omega^2$  C1  
 $0.72 mg = m \times 0.35 \omega^2$   
 $\omega = 4.49 \text{ (rad s}^{-1}\text{)}$  C1  
 $n = (\omega / 2\pi) \times 60$  B1  
 $= 43 \text{ min}^{-1} \text{ (allow 42)}$  A1 [5]
- (c) either centripetal force increases as  $r$  increases  
or centripetal force larger at edge M1  
so flies off at edge first A1 [2]  
*( $F = mr\omega^2$  so edge first – treat as special case and allow one mark)*
- 2 (a) molecule(s) rebound from wall of vessel / hits walls B1